Esra Yüksel, Ph.D.

Senior Lecturer in Theoretical Nuclear Physics

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Research Profile

My research develops microscopic nuclear theory for atomic nuclei under conditions that cannot be reproduced in the laboratory — extreme proton–neutron asymmetry near the drip lines, and the temperatures of billions of kelvin reached in core-collapse supernovae and neutron star mergers — and connects those calculations to observables in both terrestrial experiments and astrophysics.

The framework is nuclear energy density functional (EDF) theory, in both relativistic and non-relativistic formulations. A substantial part of my work has gone into constraining the functionals themselves, using nuclear ground-state properties alongside collective excitation data and, more recently, the parity-violating electron scattering measurements CREX and PREX-II; this line of work produced the DD-PCX parametrisation. In parallel I have developed finite-temperature extensions of the (quasiparticle) random phase approximation, including the charge-exchange channel, which make it possible to calculate electric and magnetic dipole strength, γ-ray strength functions, Gamow-Teller transitions, electron-capture cross sections and β-decay half-lives under stellar conditions. Applied globally, these methods showed that the limits of nuclear existence are not fixed but shift with temperature.

A second strand links finite-nuclei observables — dipole polarizability, neutron skin thickness, the symmetry energy — to neutron star radii, tidal deformabilities and compactness, bridging nuclear experiments and multimessenger astronomy. A third explores machine learning as a tool for predicting nuclear properties where experimental data run out.

I have published more than 40 peer-reviewed articles, including papers in *Nature Communications*, *Physical Review Letters*, *Physics Letters B*, *The Astrophysical Journal Letters* and *Physical Review C*.

Education and Qualifications

2024	**Fellow of the Higher Education Academy (FHEA)**, *Advanced HE* (16 October 2024)
2011–2016	**Ph.D.**, *Physics, Yildiz Technical University, Istanbul, Türkiye*
2009–2011	**M.Sc.**, *Physics, Yildiz Technical University, Istanbul, Türkiye*
2004–2009	**B.Sc.**, *Physics, Abant Izzet Baysal University, Bolu, Türkiye*

Employment

Apr 2026–Present	**Senior Lecturer**, *School of Mathematics and Physics, University of Surrey, Guildford, United Kingdom*
Oct 2022–Mar 2026	**Lecturer**, *School of Mathematics and Physics, University of Surrey, Guildford, United Kingdom*
Aug 2022	**Associate Professor**, *Department of Physics, Yildiz Technical University, Istanbul, Türkiye*
2016–2022	**Research and Teaching Assistant (Dr.)**, *Department of Physics, Yildiz Technical University, Istanbul, Türkiye*
2017–2018	**Postdoctoral Researcher**, *Department of Physics, University of Zagreb, Croatia* (14 months)
2010–2016	**Research and Teaching Assistant**, *Department of Physics, Yildiz Technical University, Istanbul, Türkiye*
2014	**Visiting Researcher**, *Institut de Physique Nucléaire d'Orsay (IPN-Orsay), France* (6 months)

Research Grants and Projects

1. **Surrey Lead** — PLANET: Physics-led Applications for Nuclear Engineering & Technology. UKRI/EPSRC Doctoral Focal Award, led by the University of York with the Universities of Cambridge, Edinburgh, Lan-

caster and Surrey. Awarded March 2026; first cohort 2026/27.

2. **Co-investigator** — TENSAR: Theory and Experiment for Nuclear Structure, Astrophysics and Reactions. Science and Technology Facilities Council (STFC), UK. 1 October 2024 – 30 September 2027.
3. **Co-investigator** — Exploring Nuclear Structure and Dynamics: From *Ab Initio* Methods to Nuclear Energy Density Functionals. DiRAC High Performance Computing Facility, UK. April 2024 – March 2027.
4. **Principal investigator** — Nuclear Structure and Dynamics under Extreme Conditions. STFC New Starter Grant ST/Y000013/1, UK. January 2023 – October 2024.
5. **Principal investigator** — The Effect of Temperature on Nuclear Structure. Supported by Higher Education Institutions, Türkiye. 2021–2022.
6. **Researcher** — Centre of Excellence for the Theory of Quantum and Complex Systems and Lie Algebra Representation (QuantiXLie), EU-supported. 9 months.
7. **Researcher** — Structure and Dynamics of Exotic Femto Systems (STRUDEF), Croatian Science Foundation. PI: Prof. Dr. Nils Paar. 14 months.
8. **Researcher** — Investigation of Nuclear Collective Modes in Neutron-Rich Nuclei with the QRPA Method. Supported by Higher Education Institutions, Türkiye. 2014–2016. PI: Asst. Prof. Dr. Kutsal Bozkurt.

Peer-Reviewed Publications

1. P. S. Koliogiannis, T. Ghosh, **E. Yüksel**, N. Paar, *Universal relation between dipole polarizability of finite nuclei and neutron-star compactness*, Phys. Rev. C **113**, 055809 (2026).
2. P. S. Koliogiannis, **E. Yüksel**, T. Ghosh, N. Paar, *Dipole polarizability of finite nuclei as a probe of neutron stars*, Astrophys. J. Lett. **996**, L18 (2026).
3. P. S. Koliogiannis, **E. Yüksel**, T. Ghosh, N. Paar, *Nuclear signatures and stellar observables: bridging terrestrial experiments and neutron star structure*, HNPS Adv. Nucl. Phys. **32**, 54–62 (2026).
4. A. Kaur, **E. Yüksel**, N. Paar, *Hot pygmy dipole strength in nickel isotopes*, Phys. Rev. C **112**, L051304 (2025). *(Letter)*
5. A. Kaur, **E. Yüksel**, N. Paar, *Electric and magnetic γ-ray strength functions at finite temperature*, Phys. Rev. C **112**, 014307 (2025).
6. M. Davies, **E. Yüksel**, J.-P. Ebran, E. Khan, P. Stevenson, *α-particle condensation in diluted ^{16}O at finite temperature*, Phys. Rev. C **112**, 014311 (2025). *(Editors' Suggestion)*
7. S. Pascu, **E. Yüksel**, et al., *Increasing octupole collectivity across the $Z = 64$ isotopic chain: $B(E3)$ values in ^{150}Gd*, Phys. Rev. Lett. **134**, 092501 (2025). [Press release]
8. J. Henderson, . . . , **E. Yüksel**, et al., *Deformation and collectivity in doubly magic ^{208}Pb*, Phys. Rev. Lett. **134**, 062502 (2025).
9. S. Pascu, **E. Yüksel**, et al., *Collective excitations in ^{150}Gd*, Phys. Rev. C **111**, 034302 (2025).
10. P. S. Koliogiannis, **E. Yüksel**, N. Paar, *Constraining neutron star properties through parity-violating electron scattering experiments and relativistic point-coupling interactions*, Phys. Lett. B **862**, 139362 (2025).
11. **E. Yüksel**, D. Soydaner, H. Bahtiyar, *Nuclear mass predictions using machine learning models*, Phys. Rev. C **109**, 064322 (2024). *(Editors' Suggestion)*
12. Abhishek, P. D. Stevenson, Y. Shi, **E. Yüksel**, A. S. Umar, *The TDHF code Sky3D version 1.2*, Comput. Phys. Commun. **301**, 109239 (2024).
13. A. Kaur, **E. Yüksel**, N. Paar, *Finite-temperature effects in magnetic dipole transitions*, Phys. Rev. C **109**, 024305 (2024).
14. A. Ravlić, **E. Yüksel**, T. Nikšić, N. Paar, *Global properties of nuclei at finite temperature within the covariant energy density functional theory*, Phys. Rev. C **109**, 014318 (2024).
15. A. Kaur, **E. Yüksel**, N. Paar, *Electric dipole transitions in the relativistic quasiparticle random-phase approximation at finite temperature*, Phys. Rev. C **109**, 014314 (2024).
16. N. Paar, **E. Yüksel**, *Nuclear energy density functionals constrained by collective nuclear excitations and parity-violating electron scattering experiments*, Nuovo Cimento C **47**, 50 (2024).
17. A. Ravlić, **E. Yüksel**, T. Nikšić, N. Paar, *Influence of the symmetry energy on the nuclear binding energies and the neutron drip line position*, Phys. Rev. C **108**, 054305 (2023).
18. A. Ravlić, **E. Yüksel**, T. Nikšić, N. Paar, *Expanding the limits of nuclear stability at finite temperature*, Nat. Commun. **14**, 4834 (2023). [Press release] [Video]
19. **E. Yüksel**, N. Paar, *Implications of parity-violating electron scattering experiments on ^{48}Ca and ^{208}Pb for nuclear energy density functionals*, Phys. Lett. B **836**, 137622 (2023).

20. **E. Yüksel**, F. Mercier, J.-P. Ebran, E. Khan, *Clustering in nuclei at finite temperature*, Phys. Rev. C **106**, 054309 (2022).
21. H. Bahtiyar, D. Soydaner, **E. Yüksel**, *Application of multilayer perceptron with data augmentation in nuclear physics*, Appl. Soft Comput. **128**, 109470 (2022).
22. A. Ravlić, **E. Yüksel**, Y. F. Niu, N. Paar, *Evolution of β-decay half-lives in stellar environments*, Phys. Rev. C **104**, 054318 (2021).
23. **E. Yüksel**, *Temperature dependence of nuclear properties: a systematic study along the isotopic and isotonic chains of nuclei*, Nucl. Phys. A **1014**, 122238 (2021).
24. **E. Yüksel**, T. Oishi, N. Paar, *Nuclear equation of state in the relativistic point-coupling model constrained by excitations in finite nuclei*, Universe **7**, 71 (2021).
25. **E. Yüksel**, D. Soydaner, H. Bahtiyar, *Nuclear binding energy predictions using neural networks: application of the multilayer perceptron*, Int. J. Mod. Phys. E **30**, 2150017 (2021).
26. A. Ravlić, **E. Yüksel**, Y. F. Niu, G. Colò, E. Khan, N. Paar, *Stellar electron-capture rates based on finite-temperature relativistic quasiparticle random-phase approximation*, Phys. Rev. C **102**, 065804 (2020).
27. **E. Yüksel**, N. Paar, G. Colò, E. Khan, Y. F. Niu, *Gamow-Teller excitations at finite temperature: competition between pairing and temperature effects*, Phys. Rev. C **101**, 044305 (2020).
28. **E. Yüksel**, T. Marketin, N. Paar, *Optimizing the relativistic energy density functional with nuclear ground state and collective excitation properties*, Phys. Rev. C **99**, 034318 (2019).
29. **E. Yüksel**, G. Colò, E. Khan, Y. F. Niu, *Nuclear excitations within microscopic EDF approaches: pairing and temperature effects on the dipole response*, Eur. Phys. J. A **55**, 230 (2019).
30. **E. Yüksel**, G. Colò, E. Khan, Y. F. Niu, *Low-energy quadrupole states in neutron-rich tin nuclei*, Phys. Rev. C **97**, 064308 (2018).
31. **E. Yüksel**, *Gamow-Teller strength in the ^{56}Fe nucleus at finite temperatures*, Turk. J. Phys. **42**, 613–620 (2018).
32. **E. Yüksel**, G. Colò, E. Khan, Y. F. Niu, K. Bozkurt, *Multipole excitations in hot nuclei within the finite-temperature quasiparticle random phase approximation framework*, Phys. Rev. C **96**, 024303 (2017).
33. **E. Yüksel**, E. Khan, K. Bozkurt, G. Colò, *Effect of temperature on the effective mass and the neutron skin of nuclei*, Eur. Phys. J. A **50**, 160 (2014).
34. **E. Yüksel**, N. Van Giai, E. Khan, K. Bozkurt, *Effects of the tensor force on the ground state and first 2^+ states of the magic ^{54}Ca nucleus*, Phys. Rev. C **89**, 064322 (2014).
35. **E. Yüksel**, E. Khan, K. Bozkurt, *Pairing properties of hot nuclei within the finite-temperature Hartree-Fock plus BCS approximation*, J. Phys. Conf. Ser. **533**, 012019 (2014).
36. **E. Yüksel**, C. Gök, K. Bozkurt, *Detailed investigation of the tensor force on the evaluation of the single-particle levels of $Z = 28$ and $Z = 82$ nuclei*, Mod. Phys. Lett. A **28**, 1350177 (2013).
37. **E. Yüksel**, E. Khan, K. Bozkurt, *The soft giant monopole resonance as a probe of the spin-orbit splitting*, Eur. Phys. J. A **49**, 124 (2013).
38. **E. Yüksel**, K. Bozkurt, *Tensor effects in dipole excitations at finite temperature*, Acta Phys. Pol. A **123**, 320–322 (2013).
39. **E. Yüksel**, E. Khan, K. Bozkurt, *Analysis of the neutron and proton contributions to the pygmy dipole mode in doubly magic nuclei*, Nucl. Phys. A **877**, 35–50 (2012).
40. **E. Yüksel**, K. Bozkurt, *Tensor effects in pygmy dipole excitation*, Int. J. Mod. Phys. E **20**, 2143–2151 (2011).

Invited Talks and Seminars

I have been invited to deliver talks and seminars at international conferences, universities and research institutes. Selected examples:

1. **E. Yüksel**, 59th Zakopane Conference on Nuclear Physics, “Extremes of the Nuclear Landscape”, Zakopane, Poland, 30 August – 6 September 2026. **Invited Lecture** (forthcoming).
2. **E. Yüksel**, *From QRPA to finite-temperature QRPA: formalism and applications*, and *Weak transitions and charge exchange at finite temperature* (focused session), ESNT workshop on (Q)RPA-based methods, CEA Saclay, France, 9 July 2026. **Invited Talks**.
3. **E. Yüksel**, *Microscopic study of dipole excitations in hot nuclei within covariant density functional theory*, 18th International Symposium on Capture Gamma-Ray Spectroscopy and Related Topics (CGS18), Monterey, California, USA, 14–19 June 2026. **Invited Plenary Talk**.
4. **E. Yüksel**, *Collective excitations in nuclei*, 8th International Conference on Collective Motion in Nuclei under Extreme Conditions (COMEX8), USA, 15–19 December 2025. **Invited Talk**.

5. **E. Yüksel**, *Constraining nuclear symmetry energy in relativistic energy density functionals: insights from CREX, PREX-II and dipole polarizability*, New Frontiers in Nuclear Physics and Nuclear Astrophysics, Ankara, Türkiye, 1–5 September 2025. **Invited Talk**.
6. **E. Yüksel**, *Nuclear landscape for hot nuclei*, 14th International Conference on Relativistic Effects in Heavy-Element Chemistry and Physics, The Netherlands, 7–11 October 2024. **Invited Talk**.
7. **E. Yüksel**, *Nuclear landscape for hot nuclei*, IJCLab, Orsay, France, 6 June 2024. **Invited Seminar**.
8. **E. Yüksel**, *Nuclear landscape for hot nuclei*, University of Manchester, United Kingdom, 15 May 2024. **Invited Seminar**.
9. **E. Yüksel**, *Implications of PREX-II and CREX experiments for relativistic nuclear energy density functionals*, University of Groningen, The Netherlands, 19 October 2023. **Invited Seminar**.
10. **E. Yüksel**, *Finite-temperature effects on nuclear excitations and weak interaction processes*, New Frontiers in Nuclear Physics and Nuclear Astrophysics, Antalya, Türkiye, 4–9 September 2023. **Invited Speaker**.
11. **E. Yüksel**, *Finite-temperature effects on nuclear excitations and weak interaction processes*, 7th International Conference on Collective Motion in Nuclei under Extreme Conditions (COMEX7), Catania, Italy, 10–16 June 2023. **Invited Speaker**.
12. **E. Yüksel**, *Constraining nuclear symmetry energy using parity-violating electron scattering experiments*, 16th International Conference on Nuclear Structure Properties (NSP), Karabük, Türkiye, 8–10 May 2023. **Invited Speaker**.
13. **E. Yüksel**, *Optimization of the relativistic energy density functionals with nuclear collective excitation properties and parity-violating electron scattering experiments*, Annual IOP Nuclear Physics Conference, University of York, United Kingdom, 4–6 April 2023. **Plenary Speaker**.
14. **E. Yüksel**, *Temperature dependence of nuclear properties and excitations*, Nuclear Theory Group, University of Zagreb, Croatia, September 2021. **Seminar**.
15. **E. Yüksel**, *Multipole excitations in nuclei at finite temperatures*, Horia Hulubei National Institute for R&D in Physics and Nuclear Engineering (ELI-NP), Bucharest, Romania, 2–9 May 2018. **Invited Seminar**.

Refereed Conference Presentations

Over 30 contributed oral and poster presentations at international and national conferences. Selected recent examples:

1. **E. Yüksel**, *Implications of PREX-II and CREX experiments for relativistic nuclear energy density functionals*, XIIth International Symposium on Nuclear Symmetry Energy (NuSYM24), Caen, France, 9–13 September 2024 (oral).
2. **E. Yüksel**, A. Kaur, N. Paar, *Electromagnetic dipole transitions in nuclei at finite temperature*, V International Conference on Nuclear Structure and Dynamics (NSD2024), Valencia, Spain, 27–31 May 2024 (oral).
3. **E. Yüksel**, *Large-scale calculation of nuclear ground-state properties*, XXI International Conference NUCLEUS-2021, St. Petersburg, Russia, 20–25 September 2021 (oral).
4. **E. Yüksel**, *Study of central depression in nucleonic densities using self-consistent mean-field models*, Turkish Physical Society 37th International Physics Congress, Muğla, Türkiye, 1–5 September 2021 (oral).
5. **E. Yüksel**, *Gamow-Teller resonances and β-decay half-lives of nuclei at finite temperature*, XIV International Conference on Nuclear Structure Properties (NSP2021), Konya, Türkiye, 2–4 June 2021 (oral).
6. T. Oishi, **E. Yüksel**, G. Kružić, N. Paar, *Nuclear symmetry energy in the relativistic mean-field model constrained by collective excitations*, NuSym2021, Catania, Italy, September–October 2021 (poster).
7. A. Ravlić, N. Paar, **E. Yüksel**, Y. F. Niu, G. Colò, E. Khan, *Description of weak-interaction rates within relativistic energy density functional theory*, 16th International Symposium on Nuclei in the Cosmos, Chengdu, China, 21–25 September 2021 (poster).
8. **E. Yüksel**, T. Marketin, N. Paar, *Implementation of nuclear excitation properties in the optimization of relativistic energy density functionals*, Turkish Physical Society 35th International Physics Congress, Muğla, Türkiye, 4–8 September 2019 (oral).
9. **E. Yüksel**, N. Paar, Y. F. Niu, *The effect of temperature on Gamow-Teller excitations in nuclei*, European Nuclear Physics Conference (EuNPC2018), Bologna, Italy, September 2018 (oral).
10. **E. Yüksel**, N. Paar, Y. F. Niu, *Gamow-Teller excitations in open-shell nuclei at finite temperatures*, Zakopane Conference on Nuclear Physics, Zakopane, Poland, August 2018 (oral).
11. **E. Yüksel**, G. Colò, E. Khan, Y. F. Niu, K. Bozkurt, *Effect of temperature on the dipole and quadrupole excitations in nuclei*, XXII Nuclear Physics Workshop, September–October 2016 (oral).
12. **E. Yüksel**, E. Khan, G. Colò, K. Bozkurt, *Evolution of multipole response in nuclei at finite temperature*, COMEX-5, Krakow, Poland, 13–15 September 2015 (oral).

13. **E. Yüksel**, E. Khan, K. Bozkurt, N. Van Giai, G. Colò, *Tensor and temperature effects on nuclear properties*, NUBA Conference Series-1, Antalya, Türkiye, 14–21 September 2014 (oral).
14. **E. Yüksel**, E. Khan, K. Bozkurt, *Pairing properties of hot nuclei within the finite-temperature Hartree-Fock plus BCS approximation*, 20th International School on Nuclear Physics, Neutron Physics and Applications, Varna, Bulgaria, 16–22 September 2013 (oral).

Supervision of Postgraduate Students

Ph.D.

- Miriam Davies — principal supervisor (October 2023 – present). Awarded a poster prize at the IOP Nuclear Physics Conference 2026 for *Alpha-particle condensation in diluted neutron-rich oxygen nuclei.*

M.Sc.

- Joshua D. Fowler (current, 2026)
- Yung-Chun Chiang (completed, 2025)
- Athina Strati (completed, 2024)
- Aloma Silva (completed, 2024)
- Kishore Thirumoorthy (completed, 2023)

In addition, I have supervised undergraduate final-year research projects.

Teaching Experience

University of Surrey, United Kingdom (2023–present)

- Module Leader: PHY1034 Essential Mathematics
- Lecturer: EEE1032 Mathematics II (Engineering Mathematics); MAT1044 Engineering Mathematics
- Essential Mathematics tutorials; small-group tutorials for Year 1 students

Yildiz Technical University, Türkiye (2010–2022)

- Quantum Physics I and II (2011–2021)
- Physics I and II (2013–2017); Physics laboratories for engineering students (2010–2020)
- Nuclear Physics I (2012–2013); Introduction to Particle Physics (2012–2013)
- Nuclear Physics Laboratory (2015–2016); Modern Physics Laboratory (2013)
- Preparation of laboratory manuals; organisation of physics examinations for engineering students

Editorial Roles, Refereeing and Review

- Subject Editor, *Proceedings of the Royal Society A* (editorial board).
- Grant reviewer for UK Research and Innovation (UKRI).
- Referee for: Physics Letters B; Physical Review C; Physical Review Letters; European Physical Journal A; Nuclear Physics A; Scientific Reports; Monthly Notices of the Royal Astronomical Society; Atomic Data and Nuclear Data Tables; Heliyon; International Journal of Modern Physics E; Indian Journal of Physics; Advances in Space Research; Particles; Universe.

Conference and Event Organisation

- Recent Developments and Challenges for (Q)RPA-Based Methods, ESNT workshop, CEA Saclay, France, 7–9 July 2026 (**Organiser**, with S. Péru, G. Colò, D. Gambacurta and E. Litvinova).
- IOP Nuclear Physics Conference 2026 (NP2026), University of Brighton, United Kingdom, 13–15 April 2026 (Organising Committee).
- XVII International Conference on Nuclear Structure Properties (NSP2025), Sivas, Türkiye, 25–27 June 2025 (Organising Committee).
- Istanbul High Energy Physics Workshop (YEFİST-2016), Istanbul, Türkiye, May 2017 (Organising Committee).
- Workshop on Nuclear Structure, Astrophysics and Reactions, Istanbul, Türkiye, May 2014 (Organising Committee).

Administrative Duties

- Surrey Lead, PLANET Doctoral Focal Award (UKRI/EPSRC), University of Surrey (2026–present)

- School Liaison Lead, School of Mathematics and Physics, University of Surrey (2024–present)
- Teaching and Marking Coordinator / Physics Department Demonstrator Coordinator, School of Mathematics and Physics, University of Surrey (2023–present)
- Mevlana Exchange Programme Committee Member, Department of Physics, Yildiz Technical University (2016–2019)
- Organiser, academic and cultural activities, Department of Physics, Yildiz Technical University (2016)

Scholarships and Fellowships

- 2219 International Postdoctoral Research Fellowship Programme, TÜBİTAK, 2018 (9 months, Croatia)
- Postdoctoral Scholarship, Official Institutions of Foreign Countries, 2017–2018 (5 months, Croatia)
- International Doctoral Research Fellowship, YÖK, 2014 (6 months, France)
- 2211-A National Ph.D. Scholarship Programme, TÜBİTAK, 2011–2016
- 2210-A National M.Sc. Scholarship Programme, TÜBİTAK, 2009–2011

Selected Media Coverage

- Surrey secures major role in three national programmes to train the next generation of the UK's nuclear workforce, University of Surrey, March 2026.
- Direct evidence revealed for rare pulsing pear shapes in gadolinium nuclei, University of Surrey, March 2025.
- "Beam me up," UK funder tells Surrey scientists, University of Surrey, April 2024.
- Groundbreaking research shows that the limits of nuclear stability change in stellar environments, University of Surrey, September 2023.

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